

## TOGUCHIN, Russian Federation

WMO: 296360

Lat: 55.2246N

Lon: 84.3859E

Elev: 173

StdP: 99.27

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -37.0      | -33.7 | -39.6                         | 0.1 | -36.8 | -36.3 | 0.1 | -33.5 | 7.4                   | -7.6 | 6.4 | -7.7 | 0.7                   | 140  | 0.586 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 29.2            | 19.0 | 27.5 | 18.4 | 26.0 | 17.8 | 20.8                | 26.2 | 19.8 | 25.2 | 18.9 | 24.3 | 2.7                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.0                            | 14.1 | 23.3 | 17.9 | 13.1 | 22.2 | 16.9 | 12.3 | 21.3 | 61.3          | 26.3 | 57.5 | 25.2 | 54.6 | 24.5 | 24.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 6.6               | 5.8  | 5.1 | -39.6                      | 31.8 | 4.5                | 1.5 | -42.8                                              | 32.8 | -45.5      | 33.7 | -48.0      | 34.5 | -51.3      | 35.6 |
|                   |      |     | -39.6                      | 22.5 | 4.3                | 1.3 | -42.7                                              | 23.4 | -45.2      | 24.2 | -47.7      | 24.9 | -50.8      | 25.8 |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 1.6    | -18.2 | -14.5 | -5.9  | 3.4   | 10.9  | 16.8  | 18.8  | 16.1  | 9.7   | 2.6   | -7.4  | -14.7 | (6)  |
| (7)  |                                                                                  | DBStd       | 14.16  | 8.94  | 7.71  | 6.13  | 5.51  | 5.27  | 4.23  | 3.16  | 3.72  | 4.40  | 5.13  | 8.07  | 8.73  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3897   | 873   | 686   | 492   | 206   | 54    | 4     | 0     | 2     | 59    | 233   | 523   | 764   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 6223   | 1132  | 919   | 750   | 447   | 234   | 76    | 33    | 89    | 259   | 487   | 773   | 1023  | (9)  |
| (10) |                                                                                  | CDD10.0     | 818    | 0     | 0     | 0     | 9     | 83    | 210   | 272   | 189   | 51    | 4     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 103    | 0     | 0     | 0     | 0     | 5     | 31    | 47    | 19    | 2     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 1082   | 0     | 0     | 0     | 3     | 109   | 337   | 407   | 199   | 27    | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 239    | 0     | 0     | 0     | 1     | 26    | 70    | 94    | 43    | 5     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.2    | 1.9   | 2.1   | 2.4   | 2.7   | 2.7   | 2.1   | 1.8   | 1.9   | 2.1   | 2.4   | 2.4   | 2.2   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 482    | 22    | 18    | 21    | 27    | 40    | 57    | 70    | 61    | 48    | 44    | 39    | 34    | (15) |
| (16) |                                                                                  | PrecMax     | 590    | 42    | 46    | 43    | 49    | 82    | 98    | 129   | 104   | 96    | 89    | 76    | 58    | (16) |
| (17) |                                                                                  | PrecMin     | 337    | 3     | 2     | 3     | 8     | 11    | 18    | 10    | 16    | 16    | 20    | 17    | 11    | (17) |
| (18) |                                                                                  | PrecStd     | 68     | 11    | 10    | 11    | 12    | 21    | 23    | 27    | 24    | 20    | 17    | 14    | 13    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 1.6   | 2.1   | 9.0   | 22.1  | 30.1  | 30.7  | 31.5  | 30.6  | 26.9  | 19.1  | 9.1   | 2.1   | (19) |
| (20) |                                                                                  |             | MCWB   | 0.2   | 0.2   | 4.3   | 12.0  | 16.1  | 18.6  | 20.0  | 19.8  | 16.3  | 11.2  | 6.9   | 0.7   | (20) |
| (21) |                                                                                  | 2%          | DB     | -1.4  | -0.1  | 6.0   | 18.5  | 26.4  | 28.6  | 29.1  | 27.7  | 22.8  | 15.7  | 6.4   | 0.2   | (21) |
| (22) |                                                                                  |             | MCWB   | -2.5  | -1.4  | 2.6   | 10.7  | 15.1  | 18.2  | 20.1  | 19.0  | 14.7  | 9.5   | 4.3   | -0.7  | (22) |
| (23) |                                                                                  | 5%          | DB     | -3.4  | -1.7  | 4.1   | 15.4  | 23.8  | 27.0  | 27.3  | 25.6  | 20.1  | 13.0  | 3.9   | -1.3  | (23) |
| (24) |                                                                                  |             | MCWB   | -4.3  | -2.9  | 1.3   | 9.0   | 13.8  | 17.7  | 19.5  | 17.9  | 13.5  | 8.4   | 2.3   | -2.2  | (24) |
| (25) |                                                                                  | 10%         | DB     | -5.4  | -3.7  | 2.4   | 12.4  | 21.1  | 25.3  | 25.6  | 23.5  | 17.5  | 10.1  | 2.0   | -3.3  | (25) |
| (26) |                                                                                  |             | MCWB   | -6.3  | -4.7  | 0.3   | 7.3   | 12.6  | 17.2  | 18.8  | 17.0  | 12.1  | 6.9   | 0.7   | -4.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 0.2   | 0.6   | 4.7   | 13.3  | 17.7  | 20.7  | 22.6  | 21.4  | 17.2  | 12.1  | 7.0   | 0.9   | (27) |
| (28) |                                                                                  |             | MCD B  | 1.4   | 1.5   | 8.1   | 19.2  | 27.2  | 26.6  | 27.8  | 27.3  | 24.0  | 17.0  | 8.2   | 1.6   | (28) |
| (29) |                                                                                  | 2%          | WB     | -2.4  | -1.4  | 2.9   | 11.5  | 16.1  | 19.6  | 21.4  | 19.7  | 15.4  | 10.2  | 4.3   | -0.5  | (29) |
| (30) |                                                                                  |             | MCD B  | -1.6  | -0.2  | 5.3   | 17.8  | 24.4  | 26.1  | 26.7  | 25.4  | 21.7  | 14.7  | 6.0   | 0.3   | (30) |
| (31) |                                                                                  | 5%          | WB     | -4.3  | -2.9  | 1.6   | 9.4   | 14.7  | 18.8  | 20.5  | 18.7  | 14.1  | 8.7   | 2.4   | -2.3  | (31) |
| (32) |                                                                                  |             | MCD B  | -3.4  | -1.7  | 3.7   | 14.8  | 21.8  | 25.4  | 25.5  | 24.5  | 19.0  | 12.4  | 3.9   | -1.4  | (32) |
| (33) |                                                                                  | 10%         | WB     | -6.3  | -4.7  | 0.3   | 7.4   | 13.3  | 17.8  | 19.5  | 17.7  | 12.7  | 7.0   | 0.8   | -4.2  | (33) |
| (34) |                                                                                  |             | MCD B  | -5.4  | -3.7  | 2.2   | 12.1  | 20.0  | 23.6  | 24.2  | 22.6  | 16.6  | 10.2  | 1.9   | -3.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 8.5    | 9.6   | 10.0  | 10.1  | 11.7  | 11.0  | 10.2  | 10.8  | 10.3  | 7.9   | 6.7   | 7.6   | (35)  |      |
| (36) |                                                                                  | 5% DB       | MCD BR | 7.7   | 7.9   | 10.5  | 15.5  | 16.5  | 14.3  | 13.2  | 14.3  | 15.4  | 13.5  | 6.9   | 7.6   | (36) |
| (37) |                                                                                  |             | MCWBR  | 7.2   | 7.1   | 7.4   | 8.8   | 7.4   | 5.9   | 5.5   | 6.4   | 7.7   | 8.1   | 5.4   | 7.3   | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 7.5   | 8.1   | 9.3   | 14.4  | 14.5  | 12.2  | 11.2  | 12.6  | 13.8  | 12.3  | 6.7   | 7.5   | (38) |
| (39) |                                                                                  |             | MCWBR  | 7.1   | 7.5   | 7.0   | 8.7   | 7.4   | 5.6   | 5.5   | 6.1   | 7.3   | 7.8   | 5.4   | 7.3   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.291  | 0.326 | 0.375 | 0.405 | 0.405 | 0.407 | 0.433 | 0.409 | 0.372 | 0.355 | 0.326 | 0.294 | (40)  |      |
| (41) |                                                                                  | taud        | 2.006  | 1.961 | 1.902 | 2.058 | 2.203 | 2.284 | 2.209 | 2.264 | 2.313 | 2.221 | 2.037 | 1.952 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 586    | 706   | 760   | 808   | 836   | 839   | 805   | 801   | 781   | 683   | 531   | 469   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 70     | 107   | 145   | 144   | 132   | 124   | 131   | 117   | 97    | 83    | 68    | 59    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.91   | 1.93  | 3.35  | 4.61  | 5.18  | 5.97  | 5.39  | 4.43  | 2.99  | 1.50  | 0.76  | 0.57  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.09   | 0.12  | 0.28  | 0.24  | 0.56  | 0.51  | 0.37  | 0.34  | 0.35  | 0.17  | 0.13  | 0.09  | (45)  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (9 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page